52. Bulgarian Mathematical Olympiad

National round, Sofia, May 17-18, 2003

Problem 1. Find the least positive integer n with the following property: if n distinct sums of the form $x_p + x_q + x_r$, $1 \leq p < q < r \leq 5$, equal 0, then $x_1 = x_2 = x_3 = x_4 = x_5 = 0$.

Problem 2. Let H be a point on the altitude CP ($P \in AB$) of an acute $\triangle ABC$. The lines AH and BH intersect BC and AC at points M and N, respectively.

a) Prove that $\sphericalangle MPC = \sphericalangle NPC$.

b) The lines MN and CP intersect at O. A line through O meets the sides of the quadrilateral $CNHM$ at points D and E. Prove that $\sphericalangle DPC = \sphericalangle EPC$.

Problem 3. Consider the sequence

$$y_1 = y_2 = 1,\ y_{n+2} = (4k-5)y_{n+1} - y_n + 4 - 2k,\ n \geq 1.$$

Find all integers k such that any term of the sequence is a perfect square.

Problem 4. A set of at least three positive integers is called *uniform* if removing any of its elements the remaining set can be disjoint into two subsets with equal sums of elements. Find the minimal cardinality of a uniform set.

Problem 5. Let a, b and c be rational numbers such that $a + b + c$ and $a^2 + b^2 + c^2$ are equal integers. Prove that the number abc can be written as a ratio of a perfect cube and a perfect square that are coprime.

Problem 6. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer n the equation $P(x) = 2^n$ has an integer solution.

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